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Walking Assistance Apparatus

Abstract

The walking assistance apparatus includes an elongated top bar having at each end of the elongated top bar a pair of vertical passageways extending through the elongated top bar. The walking assistance apparatus further includes a pair of fastener attachments positioned on each handle of an existing walker. The walking assistance apparatus further includes a pair of columns, each column is coupled to each fastener attachment, each column includes a post extending through a channel of the column and each vertical passageway of the elongated top bar. The walking assistance apparatus further includes a pair of release handles, each release handle couples to the post.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application claims the benefit of U.S. Provisional Patent Application No. 63/558,769 filed on Feb. 28, 2024.

FIELD OF THE INVENTION

[0002] The present invention relates to a walking assistance apparatus and, more particularly, to a walking assistance apparatus for stabilizing a user and relieving neck pressure of a user while utilizing a common walker.

BACKGROUND

[0003] In general, various types of devices have been employed to assist people who are recovering from various types of injuries and surgeries, or who are experiencing weakness or instability from conditions associated with advanced age or other causes, in moving. Various types of four-legged walkers have been designed. However, the various types of walkers that have been designed, fail to provide the necessary stability based on the current elevation and handle placement on the walker. Additionally, the various types of walkers that have been designed fail to relieve pressure in a user's neck which equates in some user's numbness of the hands and shortness of breath.

[0004] Accordingly, a need remains for providing the user with proper stability and relief of neck pressure while utilizing a walker. As the foregoing illustrates, the invention provides a walking assistance apparatus.

SUMMARY

[0005] The walking assistance apparatus includes an elongated top bar having at each end of the elongated top bar a pair of vertical passageways extending through the elongated top bar. The walking assistance apparatus further includes a pair of fastener attachments positioned on each handle of an existing walker. The walking assistance apparatus further includes a pair of columns, each column is coupled to each fastener attachment, each column includes a post extending through a channel of the column and each vertical passageway of the elongated top bar. The walking assistance apparatus further includes a pair of release handles, each release handle couples to the post.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0006] In the following, the present invention is described in more detail with references to the drawings in which:

[0007] FIG. 1 illustrates a perspective view of the invention positioned on a walker;

[0008] FIG. 2 illustrates a top view of an quick release handle of the invention;

[0009] FIG. 3 illustrates a right side view of the quick release handle of FIG. 2;

[0010] FIG. 4 illustrates another right side view of the quick release handle of FIG. 3;

[0011] FIG. 5 illustrates a perspective view of a fastener assembly and a clamp fastener;

[0012] FIG. 6 illustrates another perspective view of the fastener assembly and the clamp fastener;

[0013] FIG. 7 an exploded view of the invention;

[0014] FIG. 8 illustrates a top perspective view of the invention;

[0015] FIG. 9 illustrates another top perspective view of the invention;

[0016] FIG. 10 illustrates a top, front, left side perspective view of the invention;

[0017] FIG. 11 illustrates another perspective view of the invention of FIG. 10;

[0018] FIG. 12 illustrates another perspective view of the invention;

[0019] FIG. 13 illustrates another perspective view of the invention; and

[0020] FIG. 14 illustrates a perspective view of the walking assistance apparatus 1 attached to a

mobile vehicle.

DETAILED DESCRIPTION OF THE EMBODIMENT(S)

[0021] The present disclosure includes a walking assistance apparatus **1** according to the invention. In the exemplary embodiment, the walking assistance apparatus **1** generally includes an upper portion **2** and a lower portion **4**.

[0022] In an exemplary embodiment, with reference to the figures, the upper portion **2** includes a top bar **12**. The top bar **12** is an elongated cylindrical member. However, one skilled in the art will appreciate that the shape of the top bar **12** is not the exclusive shape. The top bar **12** further includes a passageway **14** extending a length of the top bar **12**. One skilled in the art would understand the applicant's design is not the exclusive embodiment.

[0023] In the exemplary embodiment, the top bar **12** further includes an inner beam **16**. The inner beam **16** is a cylindrical member which extends at least half the length of the top bar **12**. One skilled in the art would understand the applicant's design is not the exclusive embodiment.

[0024] In the exemplary embodiment, the top bar **12** further includes a plurality of passageways **22**. The at least four passageways **22** are positioned and extend through the top bar **12** in a vertical direction with respect to the top bar **12**. At least two passageways **22** are parallel, which form a continuous passageway through the top bar **12**.

[0025] In the exemplary embodiment, the inner beam **16** is positioned between the plurality of passageways **22** of the top bar **12**.

[0026] In the exemplary embodiment, the upper portion **2** further includes an extension mechanism **30**. The extension mechanism **30** includes a pair of quick release handles **32**, an adjustment cover **36** and a fastener assembly **40**.

[0027] As shown in FIG. **2**, each quick release handle **32** is a cylindrical member. One skilled in the art would understand the applicant's design is not the exclusive embodiment. In another embodiment and as shown in FIG. **4**, each quick release handle **32** is a two piece cylindrical member.

[0028] In the exemplary embodiment, the quick release handle **32** further includes a passageway **34** extending through a length of the quick release handle **32**.

[0029] In the exemplary embodiment, the adjustment cover **36** is semi cylindrical in shape and positioned between the quick release handle **32** and a fastener **44** of the fastener assembly **40**.

[0030] As shown in FIG. **3**, the fastener assembly **40** includes a threaded rod **42** and a fastener **44**. One skilled in the art would understand the applicant's design is not the exclusive embodiment. The threaded rod **42** further includes a curvature member **46**, specifically, the curvature member **46** is hook shaped. One skilled in the art would understand the applicant's design is not the exclusive embodiment.

[0031] In an exemplary embodiment and reference to the figures, the lower portion **4** includes a pair of columns **52**. Each column **52** is a cylindrical member. However, one skilled in the art will understand the shape of the column **52** is not the exclusive embodiment.

[0032] In the exemplary embodiment, each of the at least two columns **52** includes a channel **54**. The channel **54** is circular in shape and directed through both a first end **56** and a second end **58** of the column **52**.

[0033] In another exemplary embodiment, the channel **54** is threaded.

[0034] In the exemplary embodiment, the lower portion **4** further includes a fastener attachment **60**. The fastener attachment **60** is a standard hook or clasp. In the exemplary embodiment, the fastener attachment **60** is positioned around the handle of the existing walker to restrict movement of the existing walker's handles.

[0035] In another embodiment, the fastener attachment **60** is coupled to an end of each column **52**. Specifically, the second end **58** of the column. One skilled in the art would understand the applicant's design is not the exclusive embodiment.

[0036] In the exemplary embodiment, with reference to the figures, the lower portion **4** further

includes a fastener **62**. In the exemplary embodiment, the fastener **62** is a standard clamp fastener. One skilled in the art would appreciate other types of fasteners may be utilized. The fastener **62** is coupled to the fastener attachment **60**.

[0037] In the exemplary embodiment, the lower portion **4** further includes a column coupler **64**. The column coupler **64** is a fastener. One skilled in the art would understand the applicant's design is not the exclusive embodiment. In another embodiment, the column coupler **64** can be an end cap covering the passageway **22**.

[0038] In another exemplary embodiment, the securing post **70** is a threaded rod. One skilled in the art would understand the applicant's design is not the exclusive embodiment.

[0039] In another exemplary embodiment, the securing post **70** further includes a securing plate **72**. The securing plate **72** is a circular member. One skilled in the art would understand the applicant's design is not the exclusive embodiment.

[0040] In another exemplary embodiment as shown in FIGS. **10** and **11**, the walking assistance apparatus **1** further includes a fastener **80**. The fastener **80** can embody any form of fastener or clamp. One skilled in the art would understand the applicant's design is not the exclusive embodiment.

[0041] As assembled, in the exemplary embodiment the top bar **12** is positioned horizontally with respect to a level floor or ground. Each column **52** is positioned at an end of the top bar **12**. Each of the at least two columns **52** are arranged perpendicular to the top bar **12**, where the first end **56** of the column **52** is positioned under the top bar **12**. The plurality of passageways **22** are aligned with the channel **54** of the column **52**.

[0042] In the exemplary embodiment, the at least two passageways **22** of the top bar **12** align with the at least one channel **54** of the column **52**. In the exemplary embodiment, the fastener **62** tightens the fastener attachment **60** which is coupled to the existing handle of the existing walker.

[0043] In the exemplary embodiment, the securing post **70** is threaded through both a known handle of a known walker and through the first channel **54** of the column **52**. The securing post **70** is threaded through the securing plate **72** and then through the passageways **22** of the top bar **12**. Thereby, attaching/detaching the upper portion **2** and lower portion **4** to the known walker.

[0044] In another exemplary embodiment, the fastener **62** tightens the fastener attachment **60** which is coupled to the second end **58** of the column **52**. The columns **52** are then attachable/detachable to a handle of a known walker. The top bar **12** is positioned on top of each column **52** and secured by a fastener (not shown).

[0045] In another exemplary embodiment, the fastener **62** tightens the fastener attachment **60** which is coupled to the existing handle of the existing walker. The top bar **12** is positioned horizontally across the existing handles of the existing walker. The fastener **80** is positioned around both the top bar **12** and the existing handle preventing any movement.

[0046] In the exemplary embodiment, the quick release handle **32** is positioned abutting the top bar **12** and a curvature member **46** of the threaded rod **42** which couples to the securing post **70**. The threaded rod **42** is positioned through the passageway **34** of the quick release handle **32** and into the passageway **14** of the top bar **12**. The curvature member **46** of the threaded rod **42** couples to the securing post **70**. The adjustment cover **36** is positioned over the exposed portion of the threaded rod **42**. The fastener **44** is then threaded to the threaded rod **42** and tightened.

[0047] When a user is in need of additional stabilization while walking, the walking assistance apparatus **1** is utilized. The user will select a walker.

[0048] In the exemplary embodiment, the user will thread the securing post **70** through an existing pair of walker handles and position the walking assistance apparatus **1** above a pair of handles of the walker. Specifically, securing post **70** will be positioned through the second end **58** of each column **52** and into the channel **54**. Additionally, the securing post **70** is positioned through the plurality of passageways **22** of the top bar **12**. When the user needs to transport the existing walker with the walking assistance apparatus **1**, the user will include the column coupler **64** to secure the

walking assistance apparatus **1**. Additionally, the user will release the fastener **44** thereby allowing the user to release the curvature member **46** from the securing post **70** and remove the quick release handle **32** from the top bar **12** prior to transporting.

[0049] In another exemplary embodiment, the second end **58** of each of the columns **52** will be positioned on and perpendicular to the handles of the known walker. The user will then secure the walking assistance apparatus **1** to the handles by utilizing the fastener **62** of the fastener attachment **60** when circumstances need the user to generate walking in a quicker fashion.

[0050] In another exemplary embodiment, the user will position the top bar **12** over a pair of existing handles. The user will then utilize the pair of fasteners **80** to secure the top bar **12** directly to the existing walker.

[0051] It should be noted that various changes and modifications to the embodiments described herein will be apparent to those skilled in the art. Such changes and modifications may be made without departing from the spirit and scope of the present invention and without diminishing its intended advantages. For example, various embodiments of the systems and methods may be provided based on various combinations of the features and functions from the subject matter provided herein.

Claims

1. A walking assistance apparatus, comprises: an elongated top bar having at each end of the elongated top bar a pair of vertical passageways extending through the elongated top bar; a pair of fastener attachments positioned on each handle of an existing walker; a pair of columns, each column is coupled to each fastener attachment, each column includes a post extending through a channel of the column and each vertical passageway of the elongated top bar; and a pair of release handles, each release handle couples to the post.
2. The walking assistance apparatus of claim 1, wherein each release handle is a cylindrical member.
3. The walking assistance apparatus of claim 2, wherein a passageway extends through the release handle.
4. The walking assistance apparatus of claim 3, wherein the release handle further includes a rod extending through the cylindrical member.
5. The walking assistance apparatus of claim 4, wherein the rod includes a curvature member at a first end of the rod.
6. The walking assistance apparatus of claim 5, wherein the release handle further includes a cover.
7. The walking assistance apparatus of claim 6, wherein the cover is a semi cylindrical member positioned on an exposed portion of the rod.
8. The walking assistance apparatus of claim 7, wherein the release handle further includes a fastener coupled to a second end of the rod.
9. The walking assistance apparatus of claim 8, wherein the walking assistance apparatus further comprises a column coupler positioned on the post and abutting the elongated top bar.
10. The walking assistance apparatus of claim 1, wherein the post is threaded through the handle of the existing walker.
11. The walking assistance apparatus of claim 9, wherein each release handle abuts the elongated top bar.
12. The walking assistance apparatus of claim 11, wherein the curvature member couples to the post.
13. A release handle, comprises: a cylindrical member having a passageway extending through the cylindrical member; a threaded rod positioned through the passageway of the cylindrical member, the threaded rod having a first end and a second end, the first end of the threaded rod is hooked shaped; and a cover positioned over an exposed portion of the threaded rod, abutting the cylindrical

member.

14. The release handle of claim 13, wherein the release handle further includes a fastener.

15. The release handle of claim 14, wherein the fastener is positioned on the second end of the threaded rod, abutting the cover.

16. The release handle of claim 15, wherein the cover is semi cylindrical in shape.
